

MT.ST.MARYS HIGH SCHOOL KAMMENGO

PHYSICS DEPARTMENT

SENIOR THREE AOI ON REFRACTION, DISPERSION AND COLOR

TIME ALLOWED: 1 HOUR 15 MINUTES

Instructions

- Attempt **all** items and show all necessary working where applicable.

ITEM 1

A boy taking cows to graze on a hot day observed that the air above the short grass appeared like colourless flames of fire, which surprised him because he had nobody to explain it. As the cows were grazing at the water dam reservoir surrounded by high hills, he threw his grazing stick towards one of the cows, but he missed it and instead the stick dropped into the water with part of it remaining above the water surface.

When he went to pick the stick, it appeared broken, which left him surprised because water was too soft to break a stick.

The boy has come to you for assistance.

Task:

Help him to understand:

- a) the occurrences along the way to the water reservoir and at the water reservoir.
- b) the applications of the concept that caused the occurrence along the way to the water reservoir.

ITEM 2

A newly established entertainment centre in a busy trading centre began organizing outdoor music and movie nights to attract customers. To make the events more exciting, the owner installed powerful loudspeakers around the premises and colourful stage lighting for performers.

During one of the events, the performers wore costumes decorated with yellow, black and red stripes. As blue and green stage lights flashed across the stage, many spectators noticed that the colours of the costumes appeared to change. Some spectators believed that the lights were actually changing the colours of the materials.

Concerned about the complaints and the differing opinions among community members, the local council invited you to explain the observations scientifically.

Task

As a Physics learner, help the local council explain why the performers' costumes appeared to change colour when illuminated by flashing blue and green stage lights.